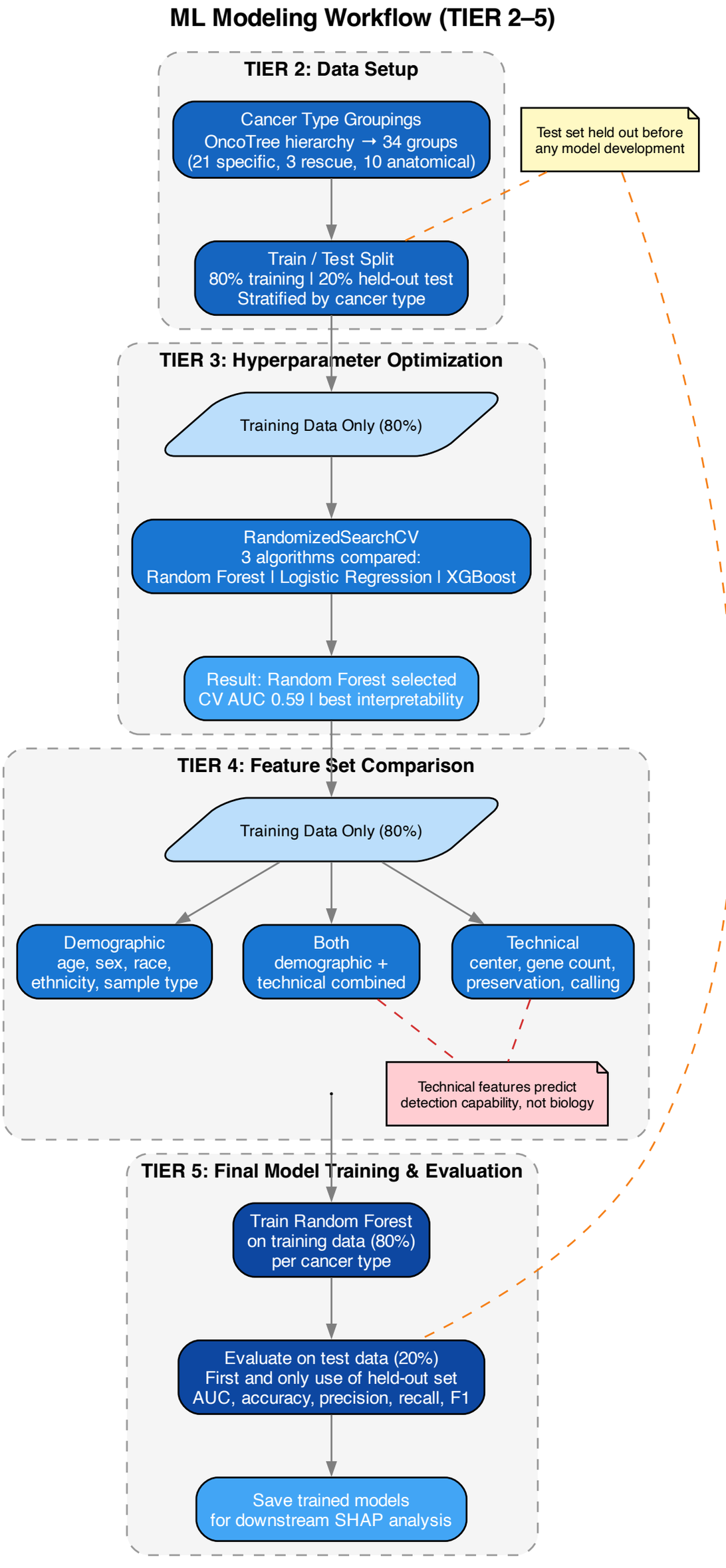

ML Modeling Workflow (TIER 2–5)
TIER 2: Data Setup
Cancer Type Groupings
OncoTree hierarchy → 34 groups
(21 specific, 3 rescue, 10 anatomical)
Test set held out before
any model development
Train / Test Split
80% training | 20% held-out test
Stratified by cancer type
TIER 3: Hyperparameter Optimization
Training Data Only (80%)
RandomizedSearchCV
3 algorithms compared:
Random Forest | Logistic Regression | XGBoost
Result: Random Forest selected
CV AUC 0.59 | best interpretability
TIER 4: Feature Set Comparison
Training Data Only (80%)
Demographic
age, sex, race,
ethnicity, sample type
Both
demographic +
technical combined
Technical
center, gene count,
preservation, calling
Technical features predict
detection capability, not biology
TIER 5: Final Model Training & Evaluation
Train Random Forest
on training data (80%)
per cancer type
Evaluate on test data (20%)
First and only use of held-out set
AUC, accuracy, precision, recall, F1
Save trained models
for downstream SHAP analysis